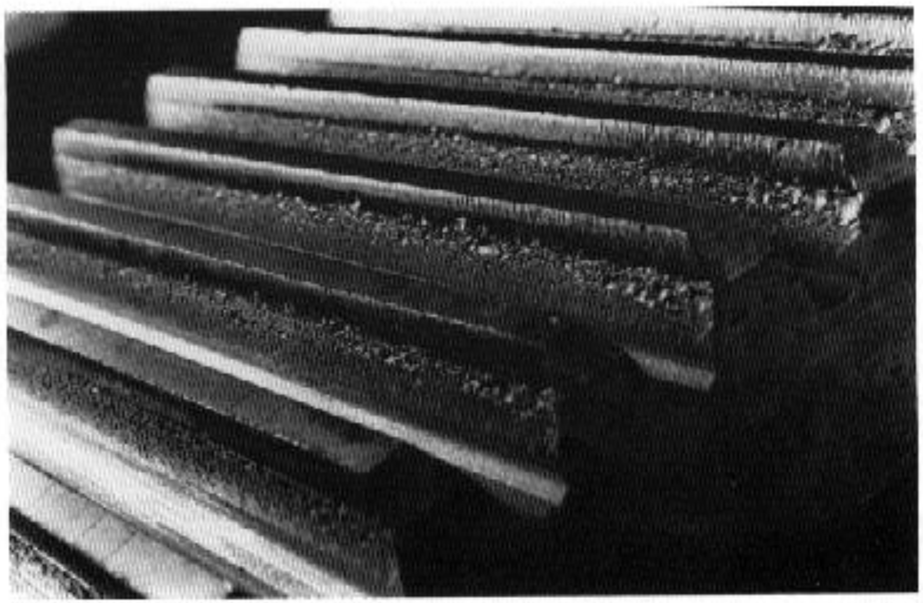
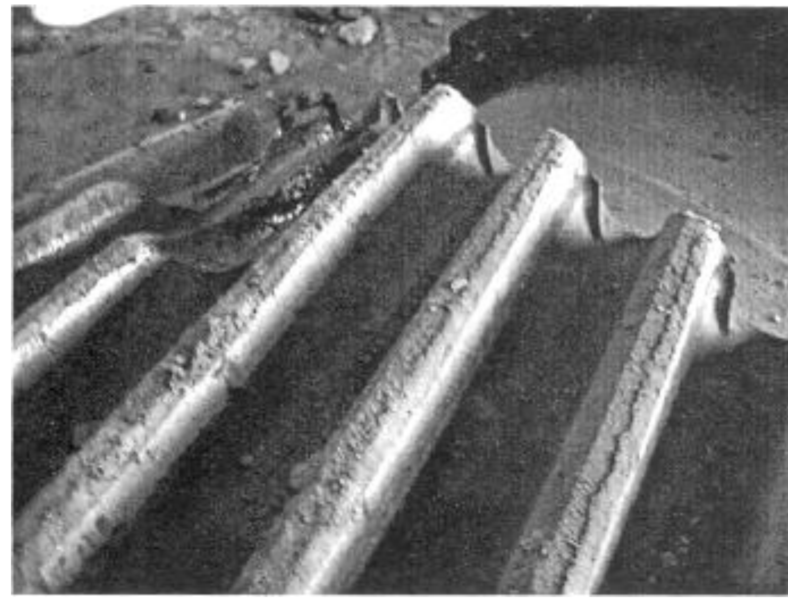
ME350 PUZZLE

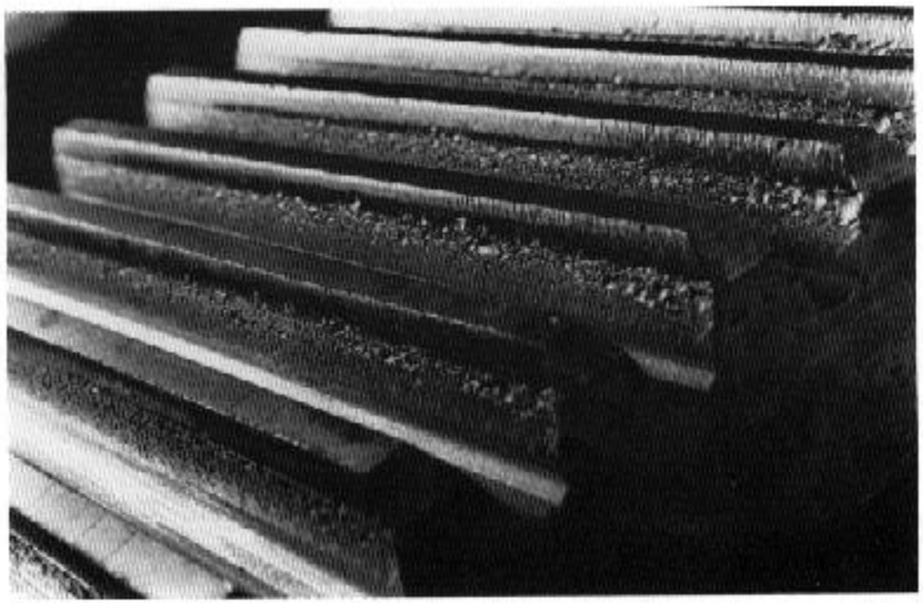


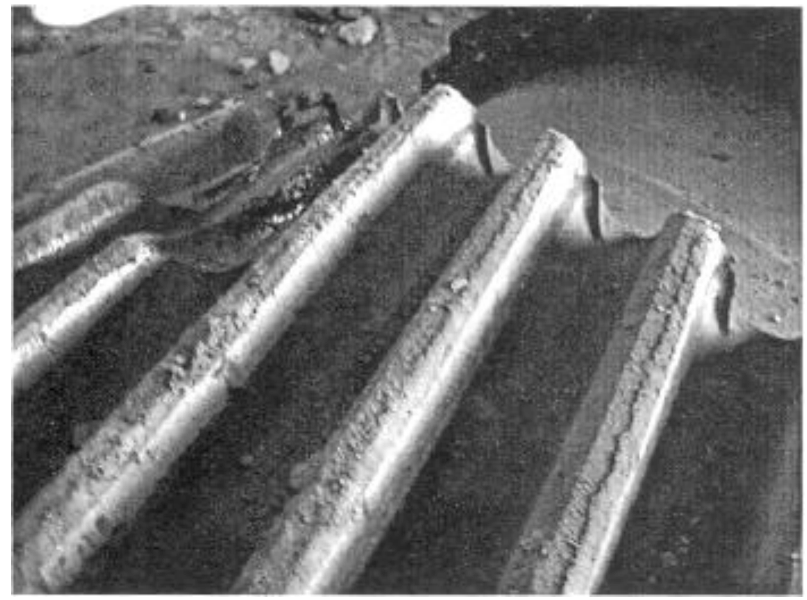


Gear teeth are typically designed to be weaker in wear than in bending. Why is this?

Images: Textron, Artec Machine

ME350 PUZZLE





Answer: Because wear occurs gradually and creates noise to give warning of deterioration, while bending causes sudden and total failure.

Images: Textron, Artec Machine

Gears – Strength: Bending Fatigue and Wear

ME 350 Lecture

Mike Umbriac

Announcements

- Statistics on the **midterm exam:**
  - **Average Score: 69.1% (33.9/49)**
  - **Standard Deviation: 13.0% (6.4/49)**
- **If you have a question about how your exam was graded,** FIRST see the posted solution and rubric on Canvas, under Files, Midterm Exam. If you still have a question after viewing the solution, you may submit a regrade request using Gradescope, within 2 weeks of the date that the exam was handed back.
- Reminder: **HW4** is due on Thursday of next week.
- Your lab sections from now until the end of the semester will be held in the **Mechatronics Lab** (1531 GGB, the north door is unlocked 8am-5pm weekdays.) Wear safety glasses when you are in this room, and work with a buddy.
- At the end of each lab session in the Mechatronics Lab, one member of each team must submit the **cleanup checklist** (online.)
- Before the next lab, please watch the **Orientation Video** (Safety Video) for the Mechatronics Lab. It is on the M.E. Student Intranet.

Mechatronics Lab

The Mechatronics Lab is located in GG Brown Laboratory room 1345 with g
hallway near the undergraduate machine shop. It is used for labs and proje
such as ME350, 450, 455 and 552. Graduate instructors (GSI) and instructior
ME350 and keycard access is typically available for ME450/455/552 students
is required of all lab users. Please review the lab rules and ask questions to
not understand something.

**** No food or drink
**** No working alone
**** No equipment leaves the lab

Mechatronics Lab Resources (Lab Rules, Safety Video, Consumable List, Eq

Gears fail when actual loads applied to teeth are greater than allowable loads.

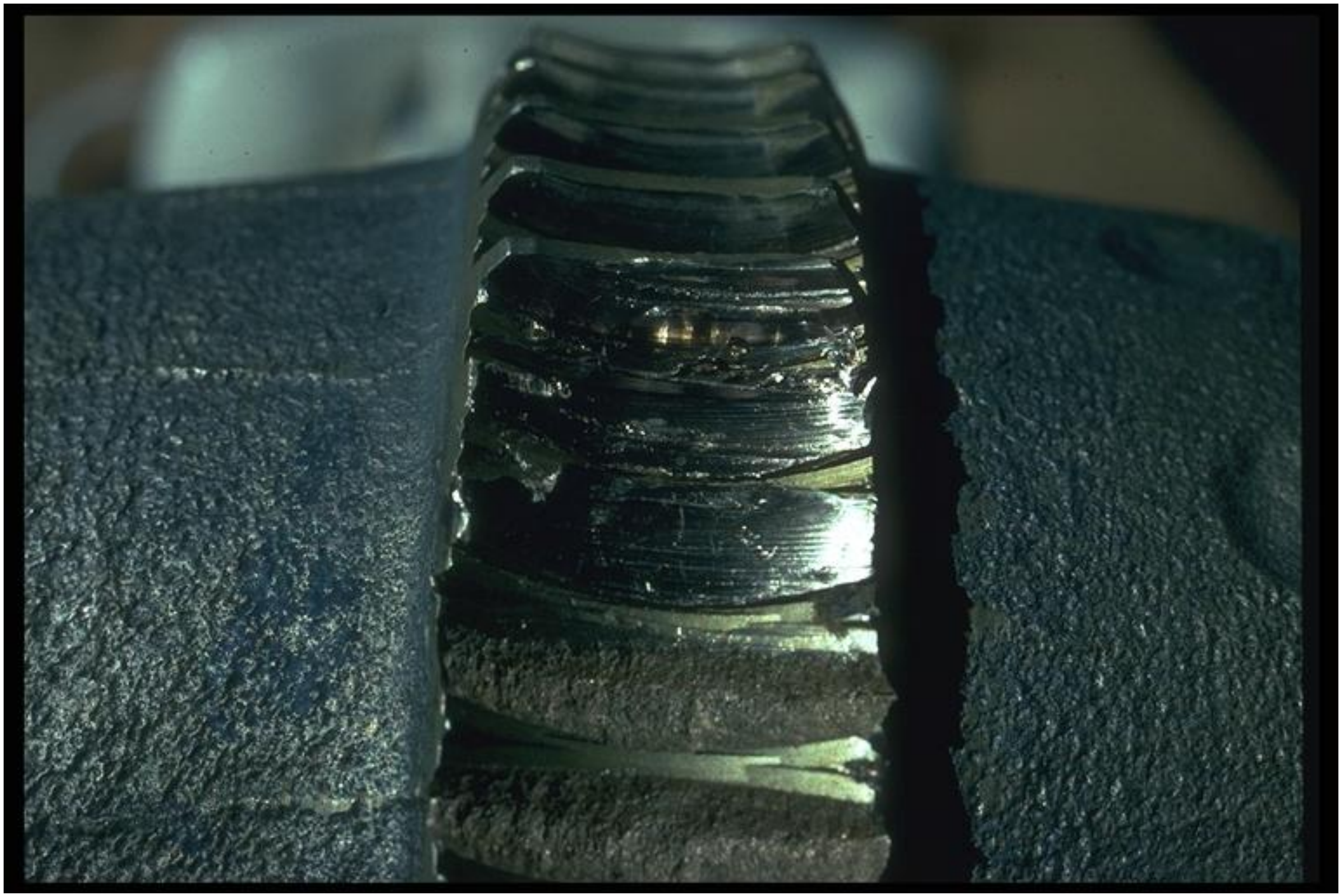
Failures:

Tooth fracture --- not strong enough

Surface failure --- wear (pitting)

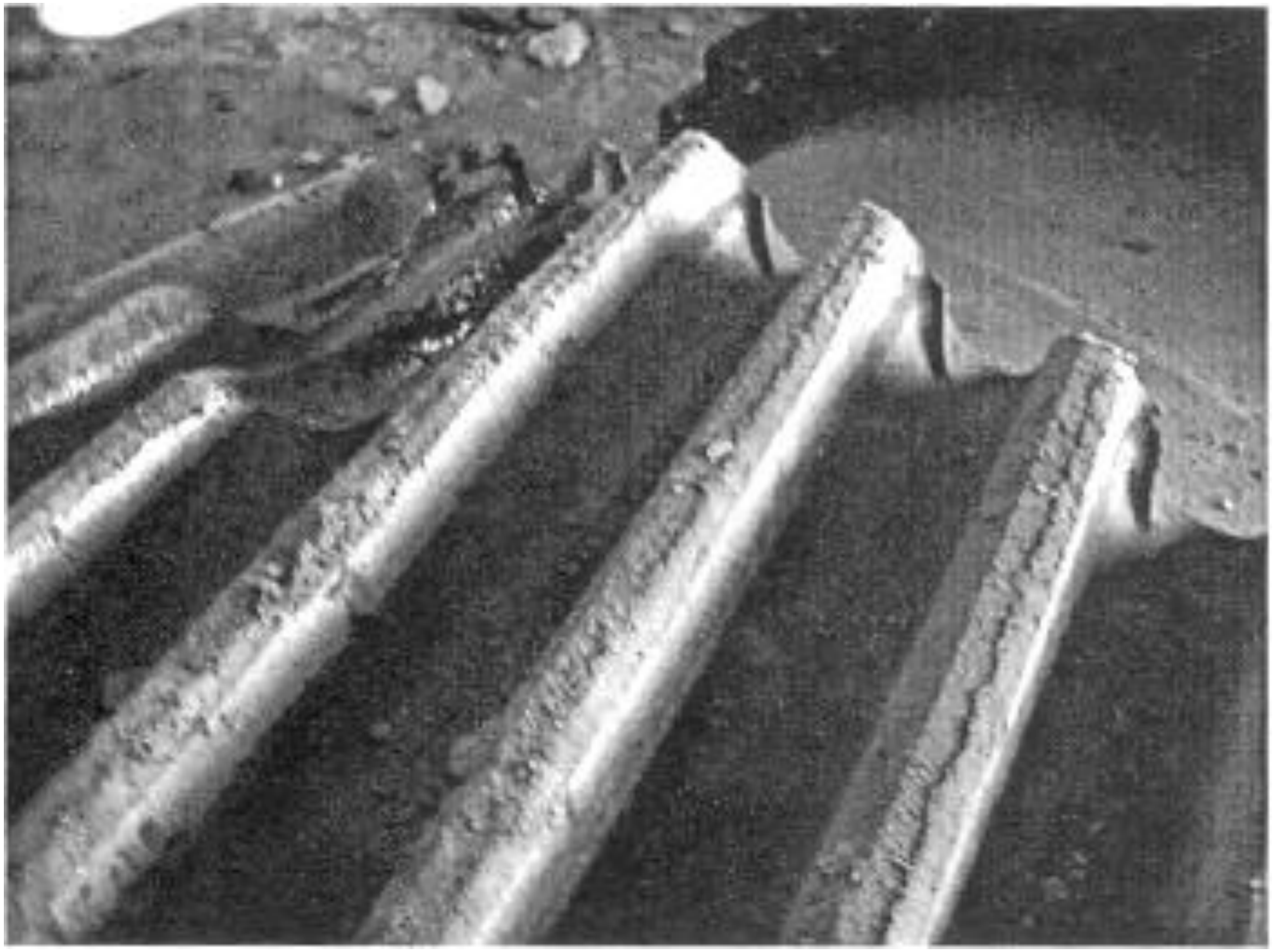
Photographs of Actual Gear Failures due to Bending Fatigue

Gear Tooth Cyclic Overload



AGMA Standard

Broken Gear Teeth on a Cement Mill

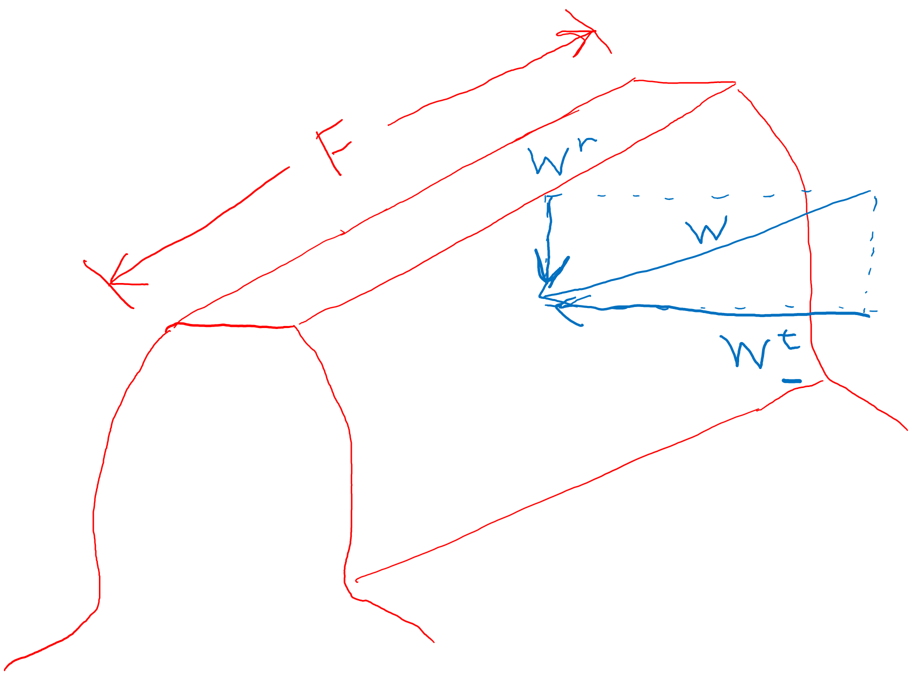


Artec-machine.com

Gear tooth failure due to misalignment of shafts

Quad4x4.com

How to model the stresses on individual gear teeth?



- Break the force into two components,
  - W^r radial force (neglect)
  - W^t **tangential force**
- Due to the complex nature of gear teeth, exact prediction of stress is difficult
- To address this, make an approximation of the tooth geometry and employ "correction factors"
- Note the "Face width", F (thickness of the gear)

Lewis Equation for Stress due to Bending (US units)

Can you guess which terms are in the numerator and denominator?

$$\frac{\qquad}{\qquad}$$

(Equation 14-7)

- Acceptable for general estimation of stresses in gear teeth
- Forms the basis for AGMA method

- σ is bending stress in psi
- K_v is the velocity factor (dimensionless)
  - As speed increases, K_v increases
- W^t is transmitted load (tangential load) in pounds
  - Can get this from eqn 13-35 in the previous lecture
- F is face width in inches
- P is diametral pitch in teeth/inch
- Y is Lewis form factor from table 14-2 (dimensionless). This one is in the denominator.

Lewis Equation for Stress due to Bending (US units)

$$\boxed{\sigma = \frac{K_v W^t P}{FY}}$$

(Equation 14-7)

- Acceptable for general estimation of stresses in gear teeth
- Forms the basis for AGMA method

- σ is bending stress in psi
- K_v is the velocity factor (dimensionless)
- W^t is transmitted load (tangential load) in pounds
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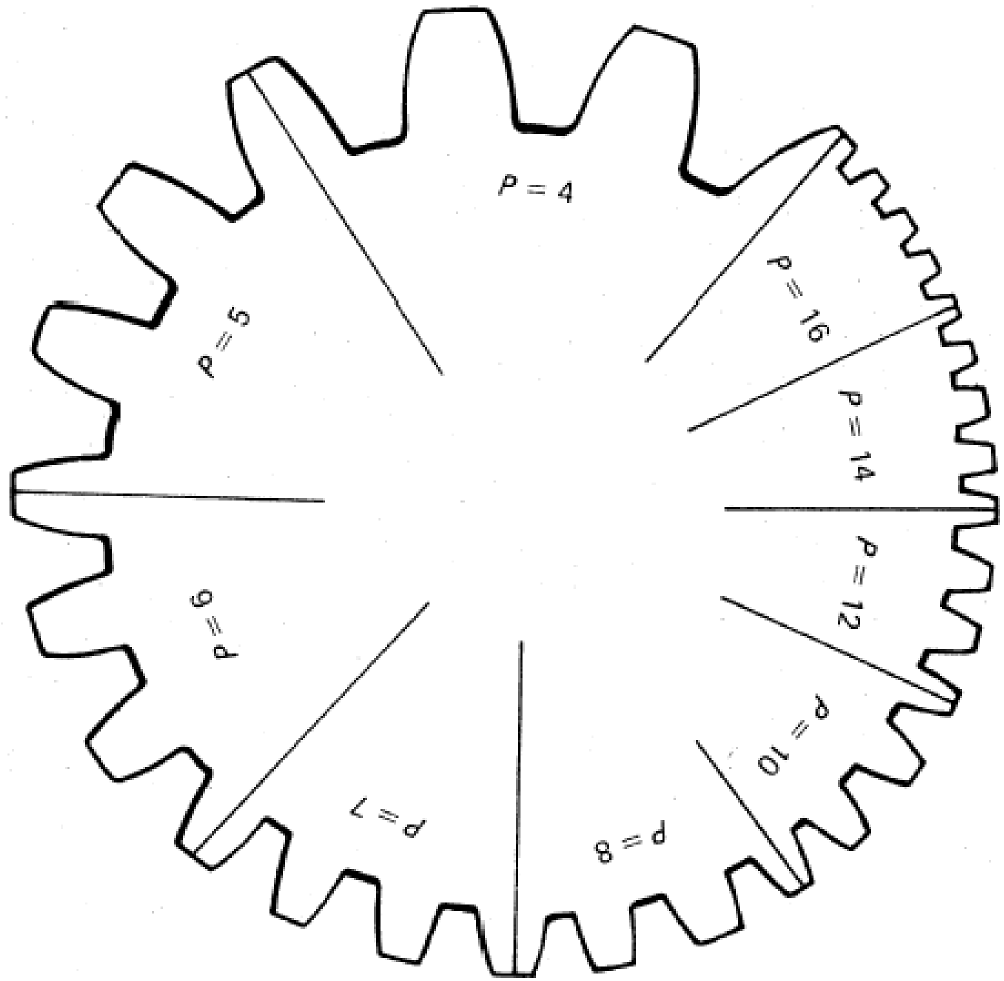
- F is face width in inches
- P is diametral pitch in teeth/inch
- Y is Lewis form factor from table 14-2 (dimensionless). This one is in the denominator.

Review of the Indexes of Tooth Size: Diametral Pitch (for U.S. units) and Module (for SI units)

$$P = \frac{N}{d}$$

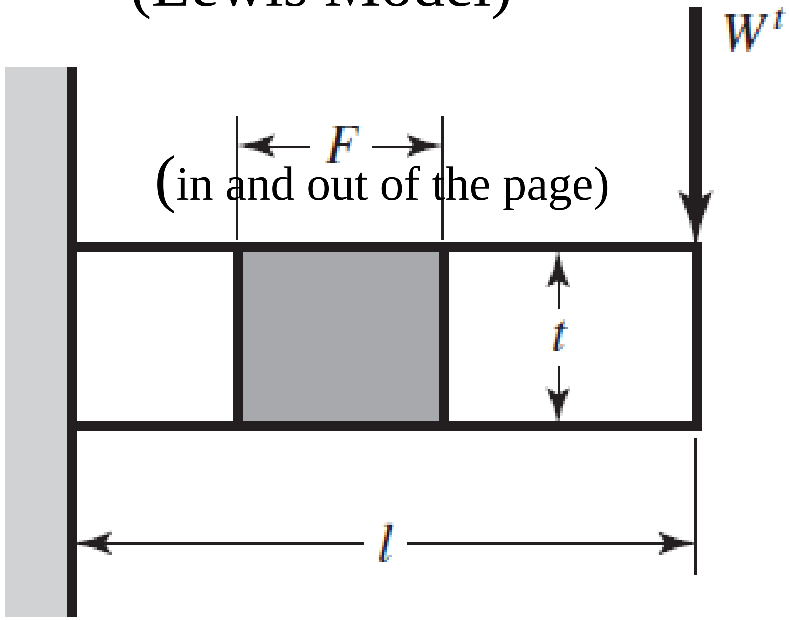
$$m = \frac{d}{N}$$



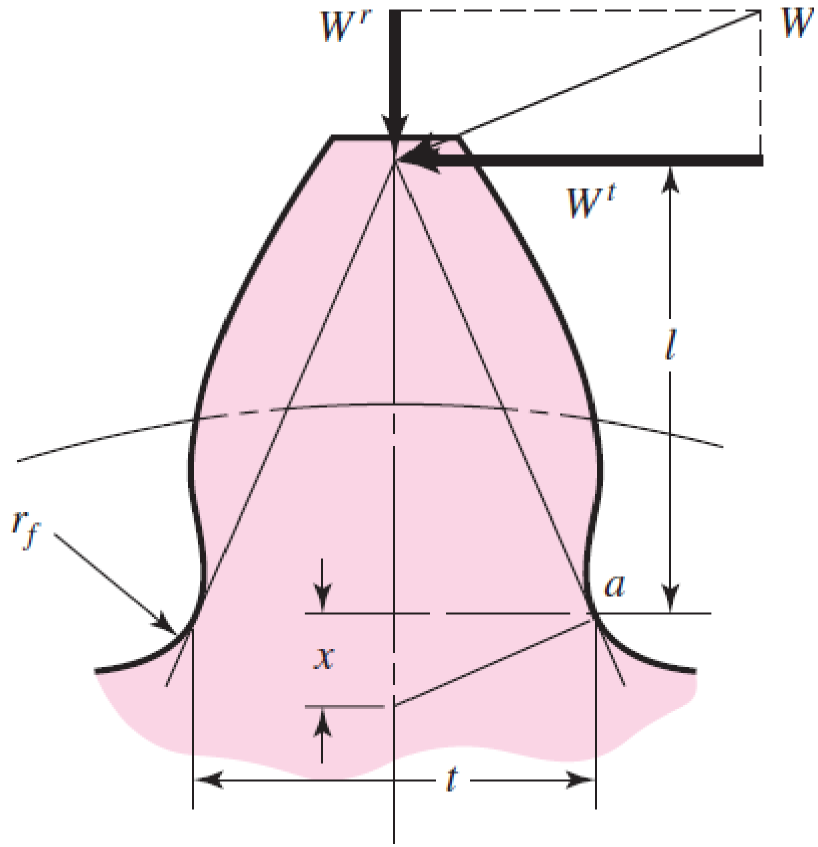
Derivation of Lewis Equation

Cantilever Beam Model of Bending Stress in Gear Tooth

(Lewis Model)



Concept: Model the gear tooth as a <u>cantilever beam</u> and use a *shape correction factor* Y to account for difference in actual tooth profile.



$$\sigma = \frac{M}{I/c} = \frac{6W^t l}{Ft^2}$$

BUT we need to apply correction factor Y.

Fig. 14–1

Values of Lewis Form Factor *Y*

(Note: This table is for 20 degree pressure angle, full-depth teeth.)

Number of Teeth	Y	Number of Teeth	Y
12	0.245	28	0.353
13	0.261	30	0.359
14	0.277	34	0.371
15	0.290	38	0.384
16	0.296	43	0.397
17	0.303	50	0.409
18	0.309	60	0.422
19	0.314	75	0.435
20	0.322	100	0.447
21	0.328	150	0.460
22	0.331	300	0.472
24	0.337	400	0.480
26	0.346	Rack	0.485

Table 14–2

Dynamic Effects: Velocity Factor K_v (US units)

- Effective load increases as velocity increases.
- *Velocity factor* K_v accounts for this. It was developed empirically.
- With pitch-line velocity V in feet per minute,

$$K_v = \frac{600 + V}{600} \qquad \text{(cast iron, cast profile)} \qquad (14\text{–}4a)$$

$$K_v = \frac{1200 + V}{1200} \qquad \text{(cut or milled profile)} \qquad (14\text{–}4b)$$

$$K_v = \frac{50 + \sqrt{V}}{50} \qquad \text{(hobbed or shaped profile)} \qquad (14\text{–}5a)$$

$$K_v = \sqrt{\frac{78 + \sqrt{V}}{78}} \qquad \text{(shaved or ground profile)} \qquad (14\text{–}5b)$$

Dynamic Effects: Velocity Factor K_v (SI units)

- With pitch-line velocity V in meters per second,

$$K_v = \frac{3.05 + V}{3.05} \qquad \text{(cast iron, cast profile)} \tag{14–6a}$$

$$K_v = \frac{6.1 + V}{6.1} \qquad \text{(cut or milled profile)} \tag{14–6b}$$

$$K_v = \frac{3.56 + \sqrt{V}}{3.56} \qquad \text{(hobbed or shaped profile)} \tag{14–6c}$$

$$K_v = \sqrt{\frac{5.56 + \sqrt{V}}{5.56}} \qquad \text{(shaved or ground profile)} \tag{14–6d}$$

Lewis Equation for Stress due to Bending (SI units)

$$\boxed{\sigma = \frac{K_v W^t}{F m Y}}$$

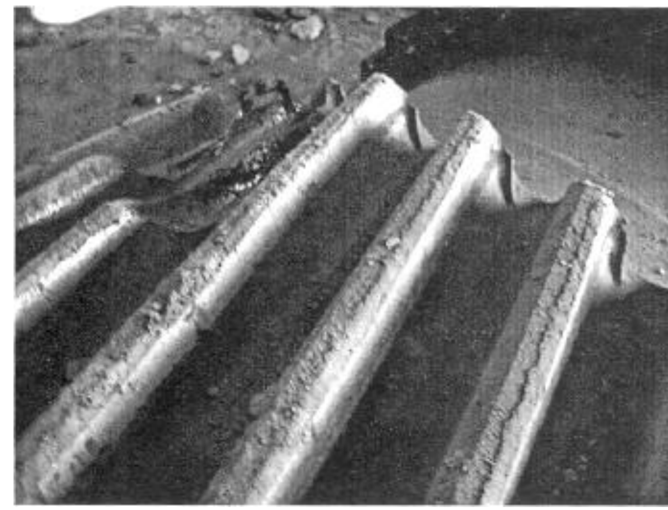
(Equation 14-8)

- σ is bending stress in MPa
- K_v is the velocity factor (dimensionless)
- W^t is transmitted load (tangential load) **in N**
  - Can get this from eqn 13-36 in the previous lecture, but be careful of units! That equation produced an answer in kN.
- F is face width in mm
- m is module in mm
- Y is Lewis form factor from table 14-2 (dimensionless)

Example Calculation

Example Problem (Problem 14-5 in book)

A steel spur pinion has a module of 1mm and 16 teeth cut on the 20 degree full-depth system and is to carry 0.15 kW at 400 rev/min.



Determine a minimum allowable face width based on an allowable bending stress of 150 MPa. (The allowable bending stress of 150 MPa in this example already accounts for a safety factor.)

Values of Lewis Form Factor *Y*

(Note: This table is for 20 degree pressure angle, full-depth teeth.)

Number of Teeth	*Y*	Number of Teeth	*Y*
12	0.245	28	0.353
13	0.261	30	0.359
14	0.277	34	0.371
15	0.290	38	0.384
16	0.296	43	0.397
17	0.303	50	0.409
18	0.309	60	0.422
19	0.314	75	0.435
20	0.322	100	0.447
21	0.328	150	0.460
22	0.331	300	0.472
24	0.337	400	0.480
26	0.346	Rack	0.485

Table 14–2

Stretch Break!

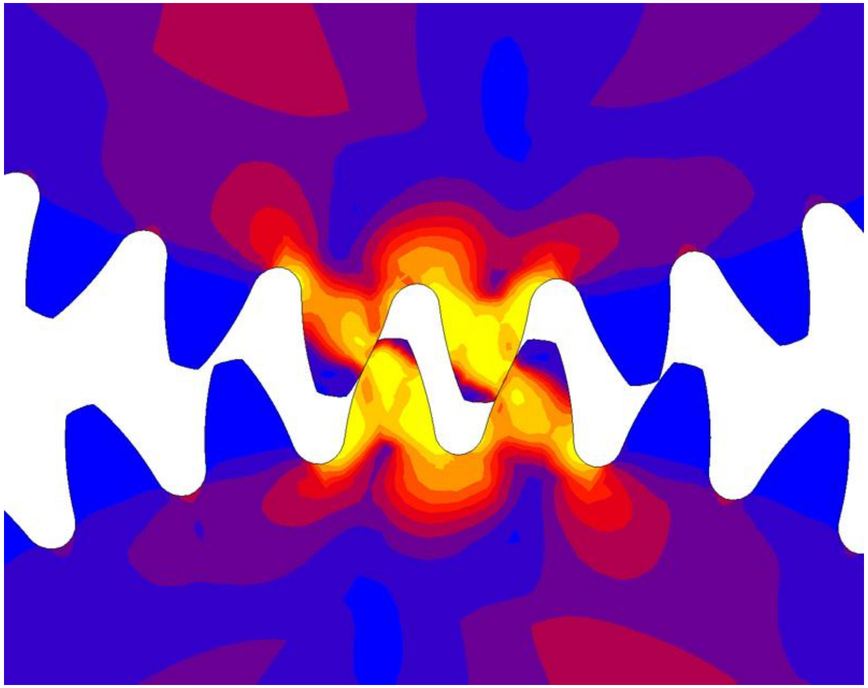
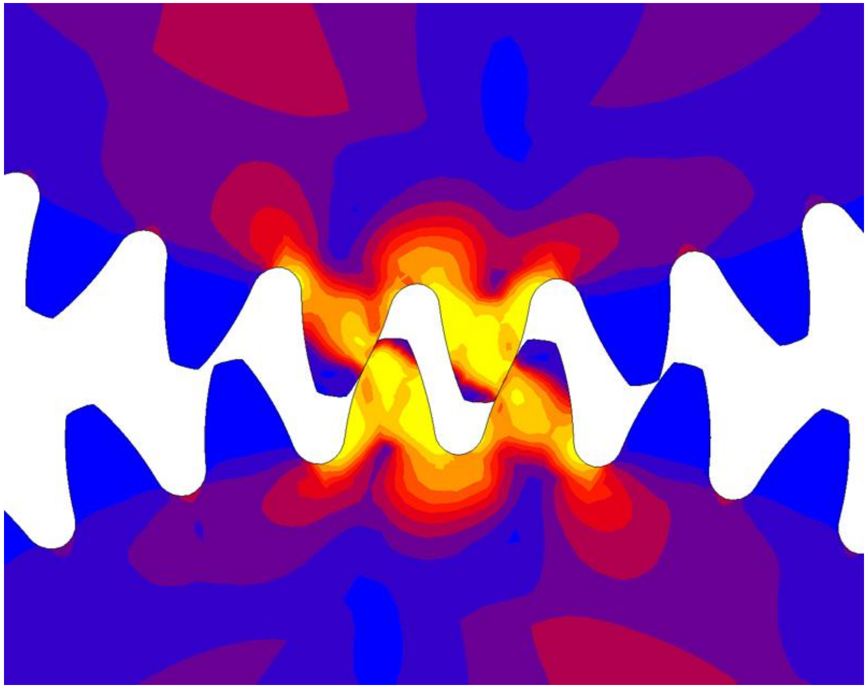


Image: mscsoftware.com

Wear

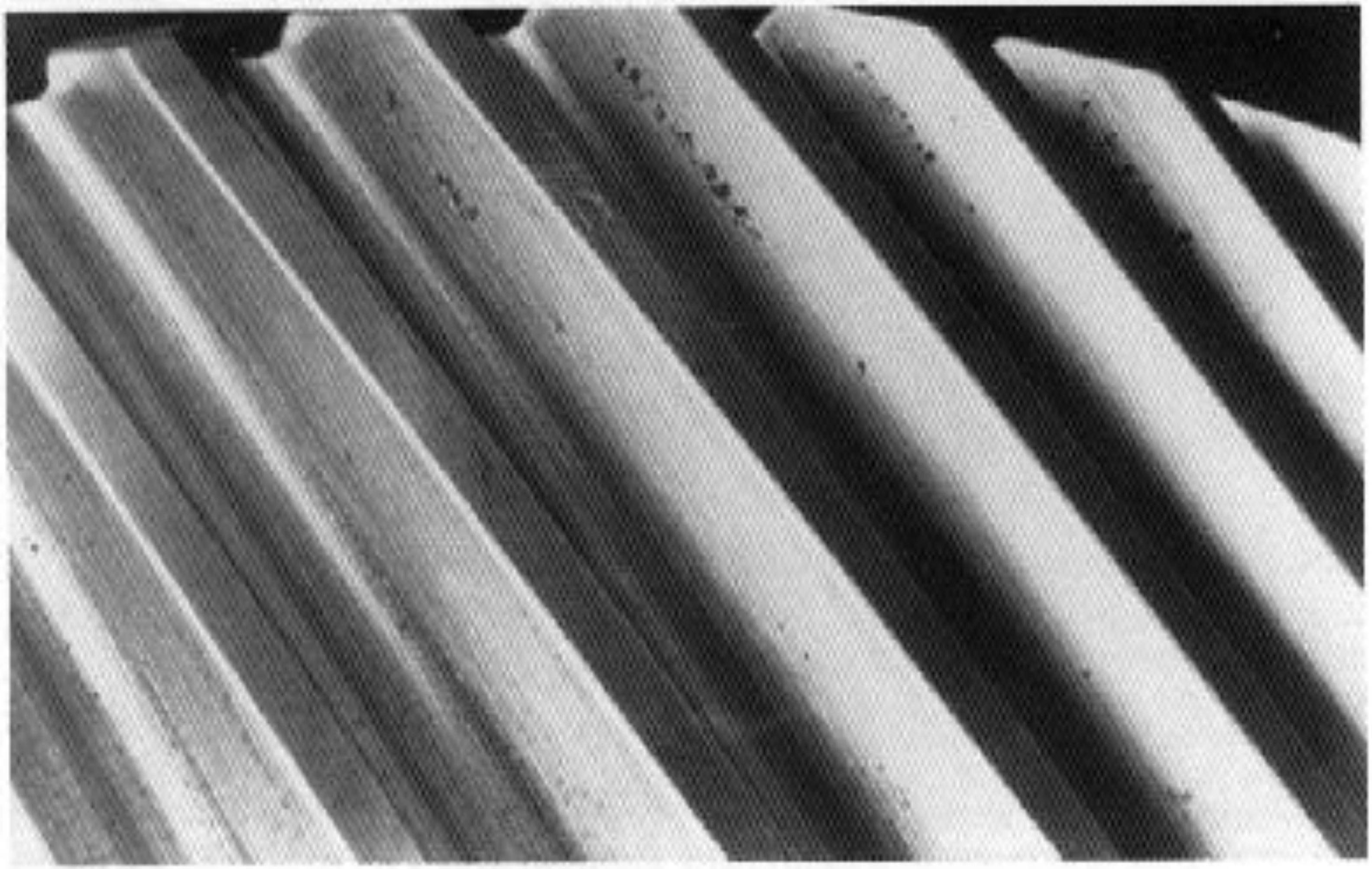
- Another failure mode for gears is **wear** due to **contact stress**.



- This usually takes the form of pitting, as is shown in the following slides.

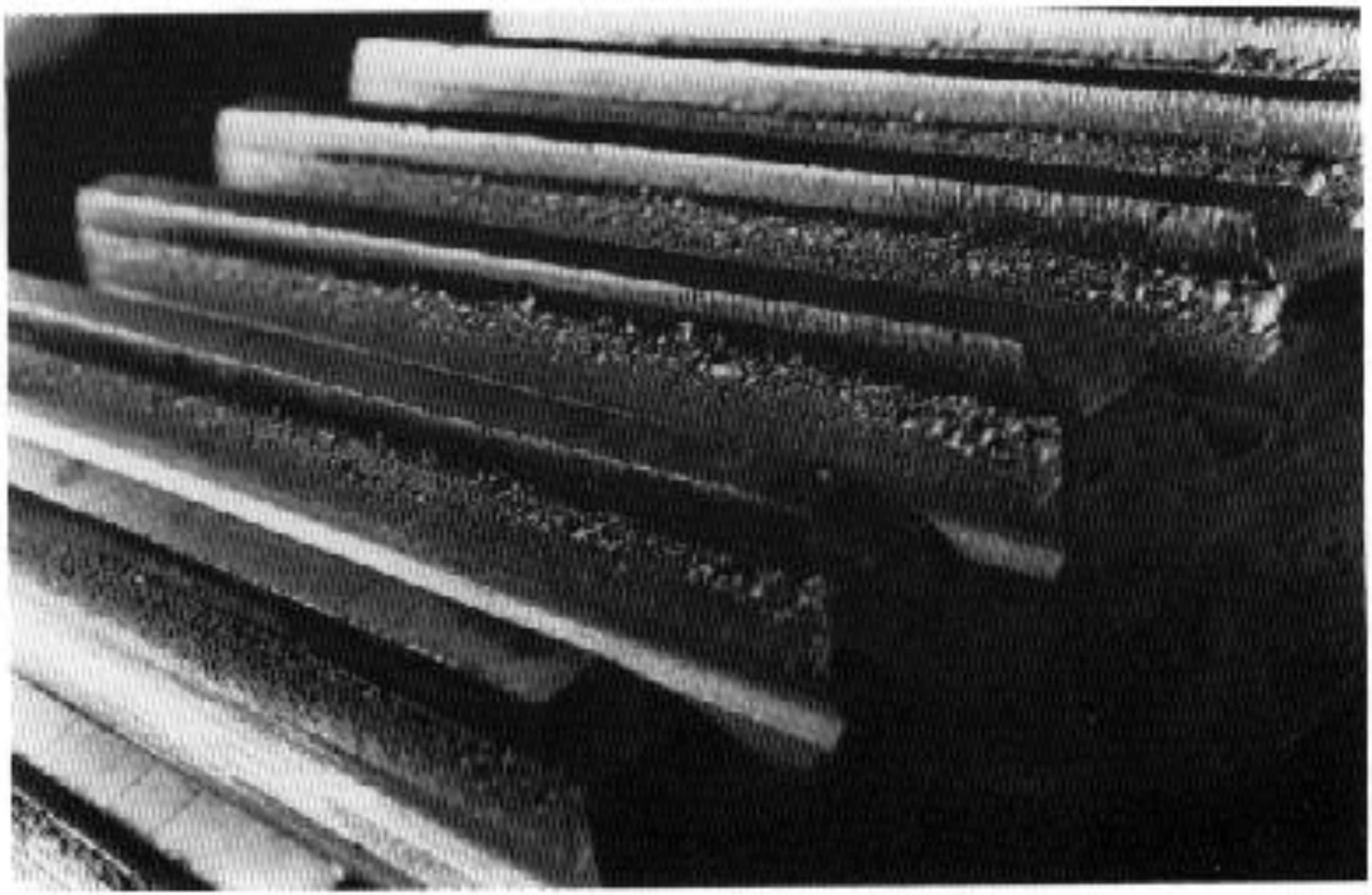
Image: mscsoftware.com

Nonprogressive Pitting (Localized)



Textron

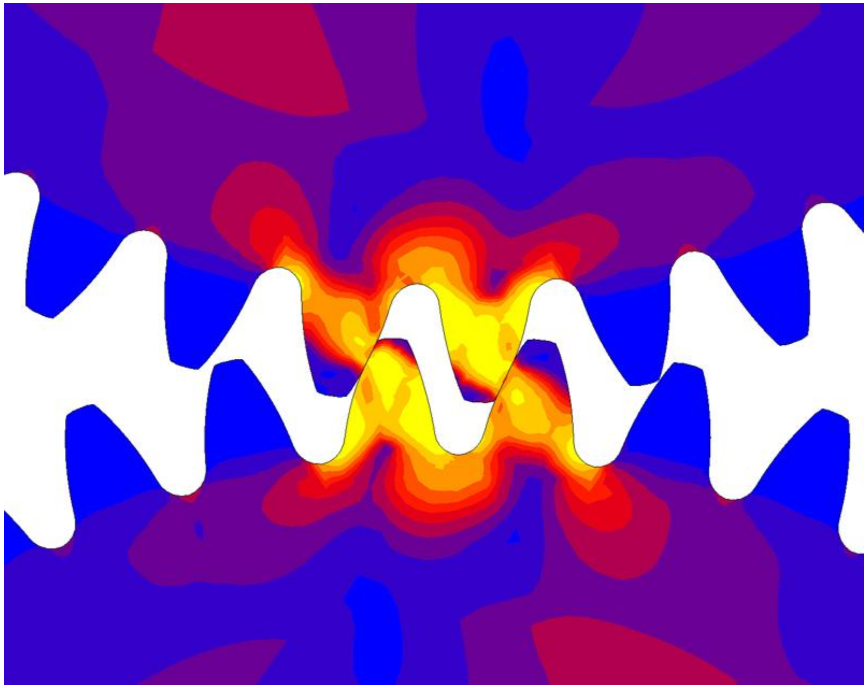
Gear Tooth Destructive Pitting



Textron

Wear

- Another failure mode for gears is **wear** due to **contact stress**.



- This usually takes the form of pitting, as is shown in the following slides.

Image: mscsoftware.com

Surface Durability (Wear)

- We model the *surface compressive stress* (*Hertzian stress*) between two cylinders (representing our gear teeth). It is based on factors including the Young's modulus E and Poisson's ratio ν.
- Critical location is usually at the pitch line, where the instantaneous radii of curvature of the tooth profiles is

$$r_1 = \frac{d_P \sin\phi}{2} \qquad r_2 = \frac{d_G \sin\phi}{2} \tag{14–12}$$

- Define *elastic coefficient* as

$$\boxed{C_p = \left[\frac{1}{\pi\left(\dfrac{1-\nu_P^2}{E_P} + \dfrac{1-\nu_G^2}{E_G}\right)}\right]^{1/2}} \tag{14–13}$$

Surface Durability (Wear)

- Incorporating elastic coefficient *Cp* and velocity factor *Kv*, the contact stress equation is

$$\boxed{\sigma_C = -C_p \left[\frac{K_v W^t}{F \cos \phi} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \right]^{1/2}} \qquad (14\text{–}14)$$

- r_1 and r_2 are given in the previous slide.
- ϕ is the pressure angle
- Again, this method is useful for estimating, and as the basis for the AGMA approach.
- NOTE: The contact stress is not linear with the transmitted load. It is directly proportional to the square root of the transmitted load.

Example #2 (based on example 14-3 in the textbook)

Given:

- Pinion material: AISI 1020 Steel
  - Young's modulus = 30 Mpsi
  - Poisson's ratio = 0.292
- Gear material: ASTM #50 Cast Iron material, but milled profile
  - Young's modulus = 14.5 Mpsi
  - Poisson's ratio = 0.211
  - Surface endurance strength S_C = 83.8 kpsi
- Pitch P = 8 teeth/in
- Number of teeth on the pinion N_P = 16
- Number of teeth on the gear N_G = 50
- Pressure angle = 20 degrees
- Speed of the pinion n_P = 1200 rpm
- Tangential Load (Transmitted Load) W^t = 382 lbs.
- Face width F = 1.5 inches (for both)

Estimate the factor of safety of the cast iron gear (which has a lower strength than the steel pinion) based on the possibility of a surface fatigue failure.

Solution

Note: In the HW and exam problems, we will provide you with the material properties E and ν and surface compressive strength S_C of the gears used.

Solution

From Table A–5 we find the elastic constants to be $E_P = 30$ Mpsi, $\nu_P = 0.292$, $E_G = 14.5$ Mpsi, $\nu_G = 0.211$. We substitute these in Eq. (14–13) to get the elastic coefficient as

$$C_p = \left\{ \frac{1}{\pi \left[\dfrac{1-(0.292)^2}{30(10^6)} + \dfrac{1-(0.211)^2}{14.5(10^6)} \right]} \right\}^{1/2} = 1817$$

From Example 14–1, the pinion pitch diameter is $d_P = 2$ in. The value for the gear is $d_G = 50/8 = 6.25$ in. Then Eq. (14–12) is used to obtain the radii of curvature at the pitch points. Thus

$$r_1 = \frac{2 \sin 20^\circ}{2} = 0.342 \text{ in} \qquad r_2 = \frac{6.25 \sin 20^\circ}{2} = 1.069 \text{ in}$$

Solution

The face width is given as $F = 1.5$ in. Use $K_v = 1.52$ from Example 14–1. Substituting all these values in Eq. (14–14) with $\phi = 20°$ gives the contact stress as

$$\sigma_C = -1817\left[\frac{1.52(380)}{1.5\cos 20°}\left(\frac{1}{0.342}+\frac{1}{1.069}\right)\right]^{1/2} = -72\,400 \text{ psi}$$

The surface endurance strength of cast iron can be estimated from

(Note: S_C is given)

$$S_C = 0.32H_B \text{ kpsi}$$

for 10^8 cycles, where S_C is in kpsi. Table A–24 gives $H_B = 262$ for ASTM No. 50 cast iron. Therefore $S_C = 0.32(262) = 83.8$ kpsi. Contact stress is not linear with transmitted load [see Eq. (14–14)]. If the factor of safety is defined as the loss-of-function load divided by the imposed load, then the ratio of loads is the ratio of stresses squared. In other words,

$$n = \frac{\text{loss-of-function load}}{\text{imposed load}} = \frac{S_C^2}{\sigma_C^2} = \left(\frac{83.8}{72.4}\right)^2 = 1.34$$

AGMA Method

- The American Gear Manufacturers Association (AGMA) provides a recommended method for gear design.
- It includes bending stress and contact stress as two failure modes.
- In each case, you compare the calculated stress to the allowable stress.
- It incorporates modifying factors to account for various situations.
- It is highly detailed, using many tables and figures.
- **We will not ask you to use this method on the exam,** but you should be aware of it for future reference. **View it as advice concerning factors that you should consider, in order to promote a long gear life in your future applications. (See the notes in red on the following slides.)**

AGMA Bending Stress

$$\sigma = \begin{cases} W^t K_o K_v K_s \dfrac{P_d}{F} \dfrac{K_m K_B}{J} & \text{(U.S. customary units)} \\ W^t K_o K_v K_s \dfrac{1}{bm_t} \dfrac{K_H K_B}{Y_J} & \text{(SI units)} \end{cases} \quad (14\text{–}15)$$

where for U.S. customary units (SI units),

- W^t is the tangential transmitted load, lbf (N)
- K_o is the overload factor
- K_v is the dynamic factor
- K_s is the size factor
- P_d is the transverse diametral pitch
- F (b) is the face width of the narrower member, in (mm)
- K_m (K_H) is the load-distribution factor
- K_B is the rim-thickness factor
- J (Y_J) is the geometry factor for bending strength (which includes root fillet stress-concentration factor K_f)
- (m_t) is the transverse metric module

Stress is increased by:

- Shock and impact
- Higher speeds
- Non-uniformity of material properties for large size gears
- Misalignment
- Thin rims for hollow gears

AGMA Allowable Bending Stress

$$\sigma_{\text{all}} = \begin{cases} \dfrac{S_t}{S_F}\dfrac{Y_N}{K_T K_R} & \text{(U.S. customary units)} \\[2ex] \dfrac{S_t}{S_F}\dfrac{Y_N}{Y_\theta Y_Z} & \text{(SI units)} \end{cases} \quad (14\text{–}17)$$

Allowable bending stress is decreased by:

- High number of cycles
- High temperatures
- High desired reliability

where for U.S. customary units (SI units),

S_t is the allowable bending stress, lbf/in^2 (N/mm^2)
Y_N is the stress cycle factor for bending stress
K_T (Y_θ) are the temperature factors
K_R (Y_Z) are the reliability factors
S_F is the AGMA factor of safety, a stress ratio

AGMA Contact Stress

$$\sigma_c = \begin{cases} C_p \sqrt{W^t K_o K_v K_s \dfrac{K_m}{d_P F} \dfrac{C_f}{I}} & \text{(U.S. customary units)} \\ Z_E \sqrt{W^t K_o K_v K_s \dfrac{K_H}{d_{w1} b} \dfrac{Z_R}{Z_I}} & \text{(SI units)} \end{cases} \qquad (14\text{–}16)$$

where W^t, K_o, K_v, K_s, K_m, F, and b are the same terms as defined for Eq. (14–15). For U.S. customary units (SI units), the additional terms are

C_p (Z_E) is an elastic coefficient, $\sqrt{\text{lbf/in}^2}$ ($\sqrt{\text{N/mm}^2}$)
C_f (Z_R) is the surface condition factor
d_P (d_{w1}) is the pitch diameter of the *pinion*, in (mm)
I (Z_I) is the geometry factor for pitting resistance

AGMA <u>Allowable</u> Contact Stress

$$\sigma_{c,\text{all}} = \begin{cases} \dfrac{S_c}{S_H} \dfrac{Z_N C_H}{K_T K_R} & \text{(U.S. customary units)} \\[2ex] \dfrac{S_c}{S_H} \dfrac{Z_N Z_W}{Y_\theta Y_Z} & \text{(SI units)} \end{cases} \tag{14–18}$$

S_c is the allowable contact stress, lbf/in^2 (N/mm^2)
Z_N is the stress cycle life factor
C_H (Z_W) are the hardness ratio factors for pitting resistance
K_T (Y_θ) are the temperature factors
K_R (Y_Z) are the reliability factors
S_H is the AGMA factor of safety, a stress ratio